Name_____ Date_____ Period_____

LESSON 3-5 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Comparing Ratios with Common Terms

Maryland Standard 6.RP.A.3b · Learning target – I can compare two ratios by scaling them until they share a common term.

REMEMBER

- To compare two ratios, scale them so one term is the **same**. Then compare the other terms.
- A bigger first number alone does **not** tell you which mix is stronger.

WORKED EXAMPLE

Punch A uses 2 cups juice to 3 cups water. Punch B uses 3 cups juice to 5 cups water. Which tastes more like juice?

Step 1 Match the juice (LCM of 2 and 3 is 6): A = 6 : 9, B = 6 : 10

Step 2 Same juice, and A has less water

Answer **Punch A tastes more like juice.**

WORD BANK

common term – the same number in both ratios

scale – multiply both numbers by the same number

stronger – more of the main thing for the same amount

PRACTICE

show your work · label units

1. Scale each ratio so the first term is 24.

Step 1 Ask: 3 × ___ = 24.

Step 2 Multiply the second number by that same number.

a. 3 : 5 = 24 : ____

b. 4 : 7 = 24 : ____

c. 6 : 11 = 24 : ____

d. 8 : 9 = 24 : ____

2. Mix P uses 3 cups blue paint to 4 cups yellow. Mix Q uses 5 cups blue to 6 cups yellow. Complete the tables. Which mix is bluer?

Step 1 Multiply 4 by 2, 3, 4 and 5 for Mix P.

Step 2 Multiply 6 by 2, 3, 4 and 5 for Mix Q.

Step 3 Look at 15 cups of blue. Less yellow means bluer.

P: blue	3	6	9	12	15
P: yellow	4				
Q: blue	5	10	15	20	25
Q: yellow	6				

Bluer mix_____

3. **Garden Seeds.** In Pack A, **9 of every 12** seeds sprout. In Pack B, **13 of every 16** seeds sprout.

Step 1 12 × 4 = 48. Multiply 9 by 4 too.

Step 2 16 × 3 = 48. Multiply 13 by 3 too.

Step 3 More seeds sprouting out of 48 is better.

a. Scale Pack A to 48 seeds: _____

b. Scale Pack B to 48 seeds: _____

c. Which pack sprouts better? _____

4. **Find the Error.** Drink X uses 4 spoons sugar to 9 cups water. Drink Y uses 6 spoons sugar to 15 cups water. Jamal says Y is sweeter because 6 > 4. Is he right? Explain.

Frame *Jamal is ___. At 12 spoons of sugar, X has ___ cups of water and Y has ___ cups, so ___ is sweeter.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Mix two glasses of a drink with different amounts. Ask someone at home which is stronger and why.

Name_____ Date_____ Period_____

LESSON 3-5 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Comparing Rates with Benchmarks

Maryland Standard 6.RP.A.3b · Learning target – I can use a benchmark to compare two rates without dividing.

REMEMBER

- A **benchmark** is a friendly amount both rates can reach.
- Scale both rates to the benchmark, then compare. Same amount: **lower price** wins.

WORKED EXAMPLE

Which is the better buy: 4 cans for $3 or 6 cans for $4.80?

Step 1 Benchmark: 12 cans (4 × 3 and 6 × 2)

Step 2 12 cans: $3 × 3 = $9.00 vs. $4.80 × 2 = $9.60

Answer **4 cans for $3 is the better buy.**

WORD BANK

benchmark – one amount you use for both choices

rate – compares two different units, like $8 for 5 pounds

better buy – costs less for the same amount

PRACTICE

show your work · label units

1. Name a benchmark amount both rates can reach.

a. a 3-pack and a 4-pack ____

b. 6 minutes and 9 minutes ____

2. Kai types 90 words in 2 minutes. Lena types 140 words in 3 minutes. Circle the faster typist.

Step 1 Kai: 2 minutes × 3 = 6 minutes. Multiply 90 by 3.

Step 2 Lena: 3 minutes × 2 = 6 minutes. Multiply 140 by 2.

	Minutes	Words	Words in 6 minutes
Kai	2	90	
Lena	3	140	

3. Bird Seed. Store A: 5 pounds of bird seed for $8. Store B: 8 pounds for $12. Use a 40-pound benchmark.

Step 1 Store A: 5 × 8 = 40. Multiply $8 by 8.

Step 2 Store B: 8 × 5 = 40. Multiply $12 by 5.

Step 3 The lower cost is the better buy.

a. Cost of 40 pounds: Store A _____ Store B _____

b. Which store is the better buy? _____

4. Trail Mix. Store C: 3 lb of trail mix for $7. Store D: 4 lb for $9. Use a 12-lb benchmark.

Step 1 3 × 4 = 12 and 4 × 3 = 12.

Step 2 Multiply each price by the same number.

a. Cost of 12 pounds: Store C _____ Store D _____

b. Which store is the better buy? _____

5. Reasoning. Car 1 goes 150 miles on 5 gallons. Car 2 goes 200 miles on 8 gallons. Which is more efficient? Use a benchmark.

Frame *At 40 gallons, Car 1 goes ___ miles and Car 2 goes ___ miles, so ___ is more efficient.*

TEST PRACTICE

choose the best answer

6. Which benchmark could you use to compare 6 bagels for $4 and 8 bagels for $5?

A 14 bagels **B** 24 bagels **C** 16 bagels **D** 10 bagels

7. Dante walks 3 mi in 50 min. Ella walks 4 mi in 60 min. Using a 12-mile benchmark, who is faster?

A Dante (200 min) **B** Ella (180 min) **C** Dante (150 min) **D** Same speed

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Compare two sizes of one item in a store ad using a benchmark.